

BIOLOGY 110: PRINCIPLES OF BIOCHEMISTRY

Lecture Notes • Cellular Respiration & ATP Synthesis

Lecture Overview

This lecture breaks down the four core stages of aerobic cellular respiration: Glycolysis, Pyruvate Oxidation, the Citric Acid (Krebs) Cycle, and Oxidative Phosphorylation. It focuses on electron carriers (NADH, FADH₂), proton gradient generation across the inner mitochondrial membrane, and ATP synthase mechanics.

Key Definitions & Core Terminology

- **Glycolysis:** Ten-step anaerobic metabolic pathway in the cytoplasm converting 1 glucose molecule into 2 pyruvate molecules, yielding net 2 ATP and 2 NADH.
- **Chemiosmosis:** The flow of protons across a selectively permeable membrane down their electrochemical gradient to drive ATP synthase.
- **Terminal Electron Acceptor:** Molecular oxygen (O₂), which accepts electrons at Complex IV of the ETC and combines with protons to form water.

Core Topics & Analytical Insights

1. Sequential Stages of Aerobic Catabolism

- Glycolysis occurs in the cytosol without oxygen requirement, yielding 2 ATP via substrate-level phosphorylation.
- Pyruvate enters mitochondria and is oxidized into Acetyl-CoA, releasing carbon dioxide and yielding NADH.
- The Citric Acid Cycle in the mitochondrial matrix yields 6 NADH, 2 FADH₂, and 2 ATP per original glucose.
- Oxidative phosphorylation in the inner membrane produces approximately 26 to 28 ATP molecules via the electron transport chain.

Important Formulas & Quantitative Facts

Overall Reaction: $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + \sim 30\text{-}32 \text{ ATP}$

Net stoichiometric chemical balance for aerobic cellular respiration.

Electron Carrier Yield: 1 NADH ~ 2.5 ATP, 1 FADH₂ ~ 1.5 ATP

Relative proton motive force contribution based on ETC entry complex.